HOMEWORK 3.9 · UNIT 3 · NC.8.EE.7

Word Problems

Oct 27 / Oct 28 · Targets: EE.7-1, EE.7-4, EE.7-5, EE.7-7, EE.7-9

Name _____ Date _____ Block _____

Remember: Word problems: 1) Name the variable. 2) Write an equation or inequality (start amount + rate × variable). 3) Solve. 4) Answer in a sentence with units. "At most" means ≤. "At least" means ≥.
Example: $20 plus $5 per hour, total $45 → $20 + 5h = 45 → 5h = 25 → h = 5$ hours

SCORE 3.0 Approaching · the basics

Targets: EE.7-1, EE.7-4, EE.7-5

1 A tablet costs $180. Rosa has $60 and saves $15 each week. Which equation tells when she can buy it? EE.7-1

(A) $60 + 15w = 180$ (B) $15 + 60w = 180$
(C) $60w + 15w = 180$ (D) $180 + 15w = 60$

2 Movie tickets cost $9 each. Dana also buys $6 of popcorn. She spends $42. How many tickets did she buy? EE.7-1

3 How many solutions? REVIEW 3.8 EE.7-4

$3x + 2 = 3x + 5$

One solution | No solution | Infinitely many

4 Four less than five times a number is 31. Find the number. EE.7-1

5 Solve. $\frac{2}{3}x + 4 = 10$ REVIEW 3.5 EE.7-5

SCORE 4.0 Proficient · on target

Targets: EE.7-5, EE.7-7

EE.7-5

6 Plan A costs $25 plus $0.10 per minute. Plan B costs $10 plus $0.15 per minute. After how many minutes do they cost the same?

EE.7-7

7 **Bri** made a mistake. Circle it. Explain it. Fix it.

BRI'S WORK

You can spend **at most** $50. Parking is $10. Tickets are $8 each.

$10 + 8t \geq 50$

$8t \geq 40$

$t \geq 5$

EE.7-7

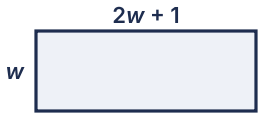
8 Marco needs **at least** 75 points. He has 33 points. Each quiz is worth 6 points. How many quizzes must he pass? Write and solve an inequality.

SCORE 5.0 Mastery · go deeper

Targets: EE.7-9

EE.7-9

9 A rectangle's length is 1 more than twice its width. The perimeter is 50. Solve $2(2w + 1) + 2w = 50$. Find the width and length.



The width is _____ and the length is _____ because _____ .

EE.7-9

10 Dana has $40. Tickets cost $6.25 each. She solved $6.25t \leq 40$ and got $t \leq 6.4$. How many tickets can she buy? Explain.

She can buy _____ tickets because _____ .